Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/16/2018
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	03/01/2016	**End Date:**	02/28/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul Ph.D. (NASA Glenn Research Center/ National Center for Space Exploration Research) Olson, Sandra Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX16AE84A		
Performance Goal No.:			
Performance Goal Text:			

NOTE: Continuation of "Flame spread and extinction over solid materials," grant NNX10AD99A.
Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives.

This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a

Task Description:	representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, shoulder and wake regions result in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with several imbedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	The PhD dissertation by Michael Johnston summarized the work performed recently. This is broken into multiple related sections culminating in support of the space flight experiment Growth and Extinction Limits (GEL). GEL is a solid fuel combustion experiment in microgravity which intends to study the significant effect of solid fuel preheating on ignitability, flammability, flame propagation, and extinction. It is part of the combined Solid Flammability, Ignitability, and Extinction (SoFIE) project which is planned to operate in the Combustion Integrated Rack (CIR) aboard the International Space Station (ISS). As part of this work, 1. A ground based test device was designed and constructed to measure the real time burning rate of solid fuels through gravimetric analysis within an upward facing low speed variable oxygen tunnel, similar to a commercial limiting oxygen index device. This device is described in detail in Johnston's PhD dissertation and journal publication #2. 2. The validation of the above device using ethanol tea lamps was presented. 3. Direct measurement of the local burning rate in the forward stagnation region of thick polymethyl methacrylate spherical shells in a low stretch flame configuration (similar to those expected in the GEL flight experiment) was presented as a function of heat loss to the solid and gas feed oxygen concentration. The existence of minimum critical local burning rate as a property of the material alone had been suggested by previous works due to the desire for simplification in crude numerical models. This was not the case, and burning rate was found to be a parameter of flame heat loss to the solid and ambient oxygen concentration. 4. The design and construction of spherical PMMA samples for the GEL flight experiment, with thermocouples embedded into the solid interior was discussed. 5. A ground testing campaign of GEL sample prototypes in the NASA Zero-G Facility was discussed as the flame transitions from 1G to 0G near the low oxygen burning limit with transition triggered when a prescribed internal temperature profile was reached. An approximate extinction boundary was determined using this procedure. 6. The design and construction considerations for the Growth and Extinction Limits experimental apparatus in ISS were discussed, including the proposed test matrix, igniter testing, and gaseous oxidizer resource consumption. Journal Publication: 1. Zhao, X., Liao, Y.-T. T., Johnston, M.C., T'ien, J.S., Ferkul, P.V. and Olson, S.L., Concurrent flame growth, spread and quenching over composite fabric samples in low speed purely forced flow in microgravity, Proceedings of the Combustion Institute 36 (2017) 2971–2978. http://dx.doi.org/ (Ed. note 5/17/18: included with 2017 report Bibliography) 2. Johnston, M.C. and T'ien, J.S., Gravimetric measurement of solid and liquid fuel burning rate near and at the low oxygen limit, Fire Safety Journal, 91, July 2017, Pages 140-146. Conference presentation: 1. M.C. Johnston, J.S. T'ien, Gravimetric measurement of solid and liquid fuel burning rate near and at the low oxygen extinction limit. 12th International Symposium of the International Association for Fire Safety Science, June 12-16th, 2017, Lund, Sweden 2. M.C. Johnston, S.L. Olson, P.V. Ferkul, J.S. T'ien, Preheating effects on the flammability of a thick fuel in low-stretch flow in microgravity, 33rd Annual Meeting of the American Society of Gravitational and Space Research, Oct. 25-28, 2017, Renton, WA. PhD dissertation: Michael C. Johnston, "Growth and Extinction Limits: Ground Based Testing of Solid Fuel Combustion in Low Stretch Conditions in Support of Space Flight Experiments", PhD dissertation, Case Western Reserve University, January, 2018.
Bibliography Type:	Description: (Last Updated: 09/19/2025)

Abstracts for Journals and Proceedings	Johnston MC, Olson SL, Ferkul PV, T'ien JS. "Preheating effects on the flammability of a thick fuel in low-stretch flow in microgravity." Abstract and presentation. 33rd Annual Meeting of the American Society of Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society of Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Johnston MC, T'ien JS. "Gravimetric measurement of solid and liquid fuel burning rate near and at the low oxygen extinction limit." Fire Safety Journal. 2017 Jul;91:140-6. https://doi.org/10.1016/j.firesaf.2017.03.027 , Jul-2017
Dissertations and Theses	Johnston MC. (Michael C. Johnston) "Growth and Extinction Limits: Ground Based Testing of Solid Fuel Combustion in Low Stretch Conditions in Support of Space Flight Experiments." PhD dissertation, Case Western Reserve University, January 2018. http://rave.ohiolink.edu/etdc/view?acc_num=case1511915506999995 , Jan-2018